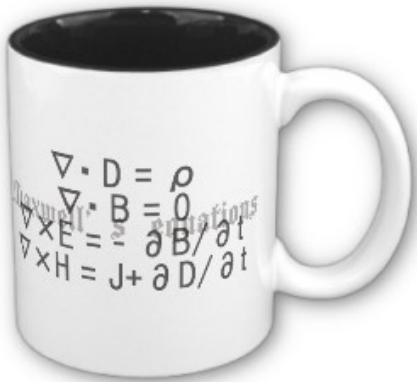


ECE 3317

Dr. Jiefu Chen

Notes 4

Maxwell's Equations



Adapted from notes by Prof. Stuart A. Long

Electromagnetic Fields

Four vector quantities

E electric field strength [Volt/meter]

D electric flux density [Coulomb/meter²]

H magnetic field strength [Amp/meter]

B magnetic flux density [Weber/meter²] or [Tesla]



each are functions of space and time
e.g. $E(x,y,z,t)$

J electric current density [Amp/meter²]

ρ_v electric charge density [Coul/meter³]



Sources generating
electromagnetic fields

Maxwell's Equations

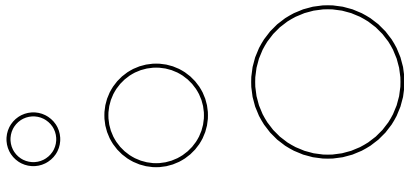
(Time-varying, differential form)

$$\nabla \times \underline{E} = - \frac{\partial \underline{B}}{\partial t}$$

$$\nabla \times \underline{H} = \underline{J} + \frac{\partial \underline{D}}{\partial t}$$

$$\nabla \cdot \underline{B} = 0$$

$$\nabla \cdot \underline{D} = \rho_v$$



James Clerk Maxwell.

Maxwell

James Clerk Maxwell (1831–1879)

James Clerk Maxwell was a Scottish mathematician and theoretical physicist. His most significant achievement was the development of the classical electromagnetic theory, synthesizing all previous unrelated observations, experiments and equations of electricity, magnetism and even optics into a consistent theory. His set of equations—Maxwell's equations—demonstrated that electricity, magnetism and even light are all manifestations of the same phenomenon: the electromagnetic field. From that moment on, all other classical laws or equations of these disciplines became simplified cases of Maxwell's equations. Maxwell's work in electromagnetism has been called the "*second great unification in physics*", after the first one carried out by Isaac Newton.

Maxwell demonstrated that electric and magnetic fields travel through space in the form of waves, and at the constant speed of light. Finally, in 1864 Maxwell wrote *A Dynamical Theory of the Electromagnetic Field* where he first proposed that light was in fact undulations in the same medium that is the cause of electric and magnetic phenomena. His work in producing a unified model of electromagnetism is considered to be one of the greatest advances in physics.



James Clerk Maxwell.

(Wikipedia)

Maxwell's Equations (cont.)

$$\nabla \times \underline{E} = - \frac{\partial \underline{B}}{\partial t} \quad \text{Faraday's law}$$

$$\nabla \times \underline{H} = \underline{J} + \frac{\partial \underline{D}}{\partial t} \quad \text{Ampere's law}$$

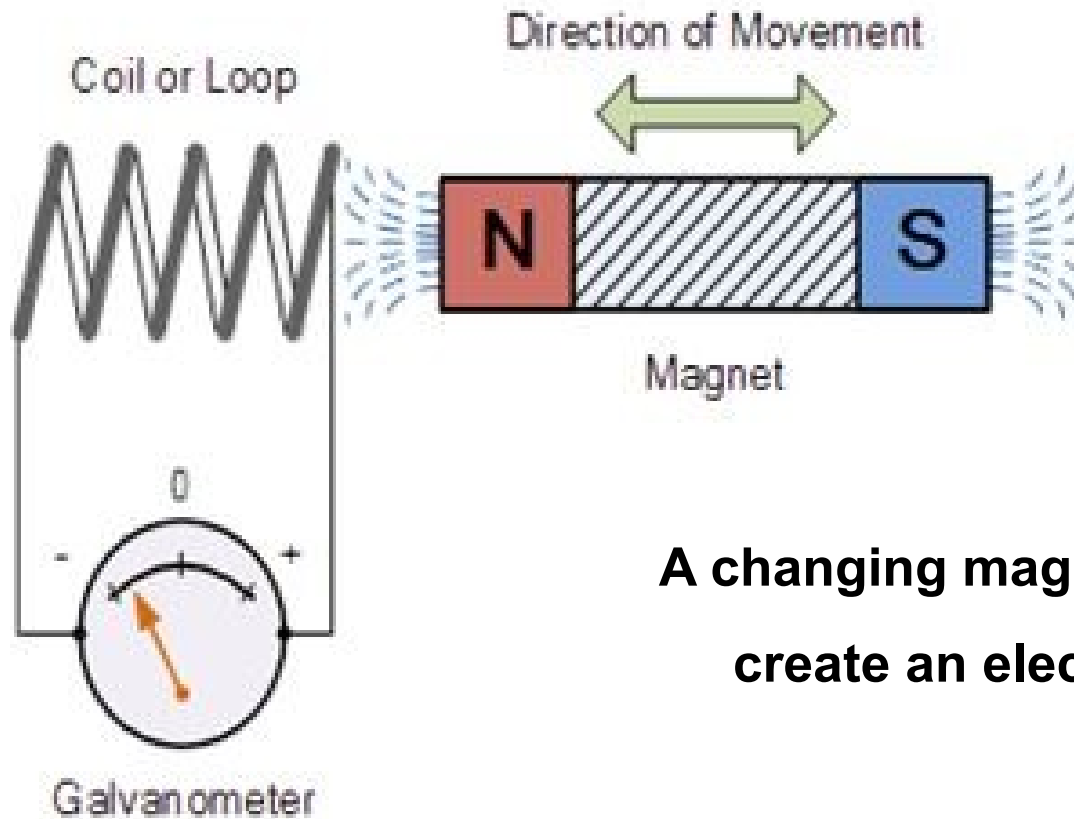
$$\nabla \cdot \underline{B} = 0 \quad \text{Magnetic Gauss law}$$

$$\nabla \cdot \underline{D} = \rho_v \quad \text{Electric Gauss law}$$

Faraday's Law

$$\nabla \times \underline{E} = - \frac{\partial \underline{B}}{\partial t}$$

www.electrical4u.com

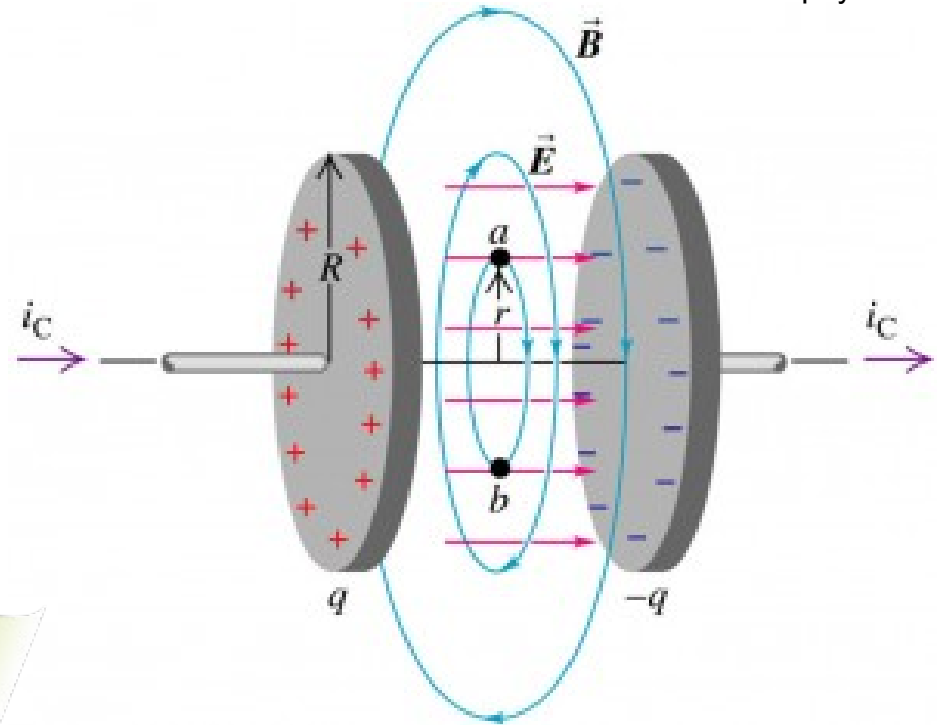
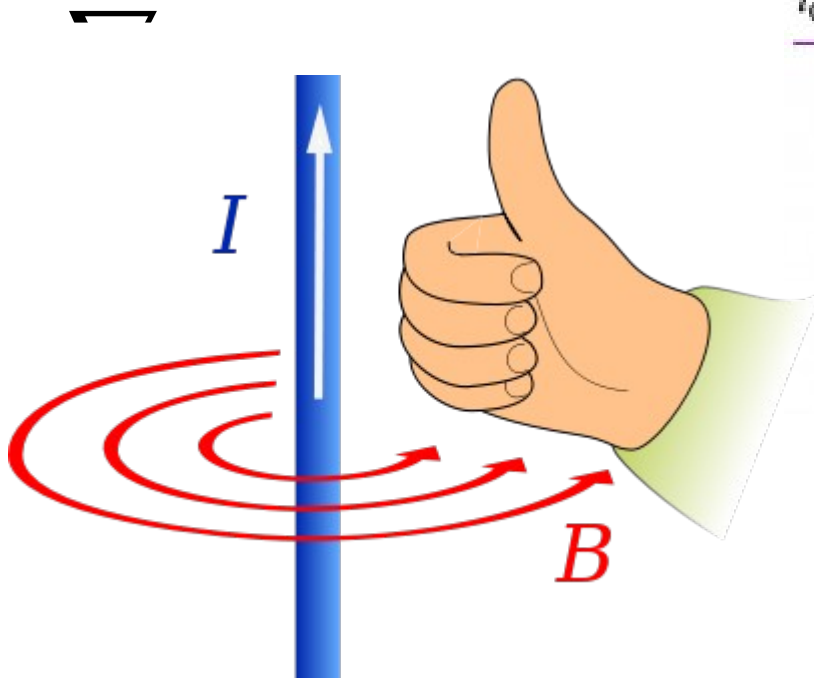


**A changing magnetic field can
create an electric current**

Ampere's Law

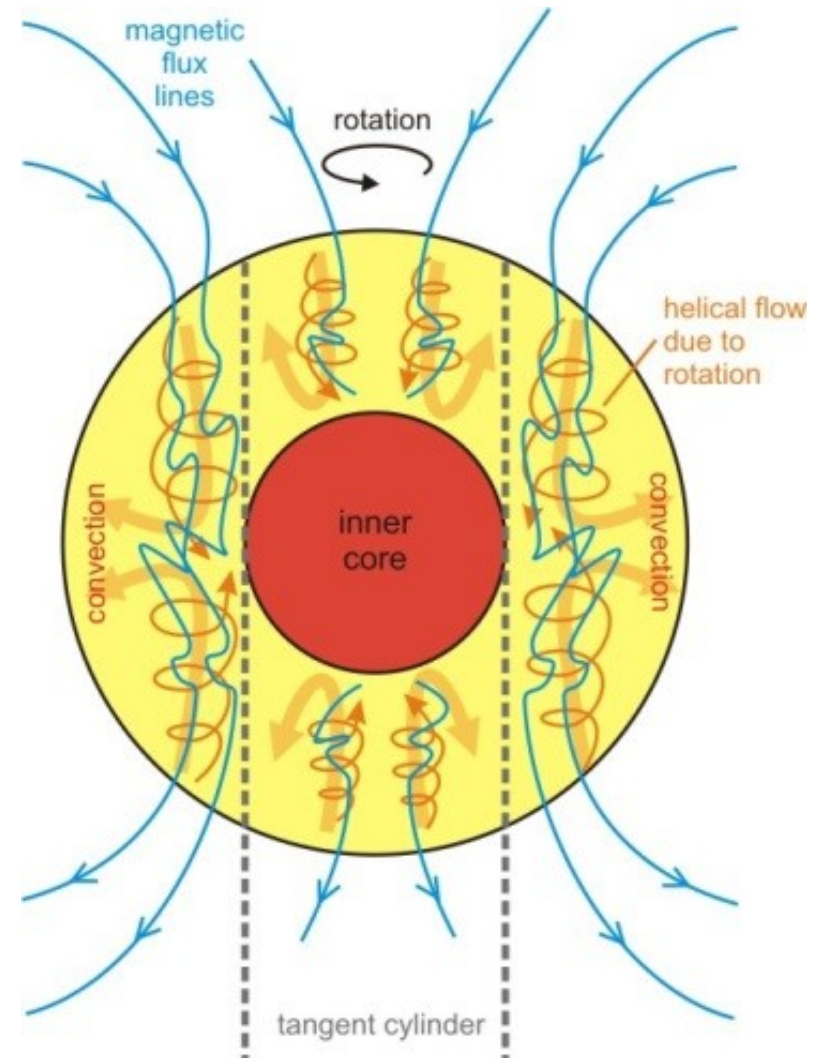
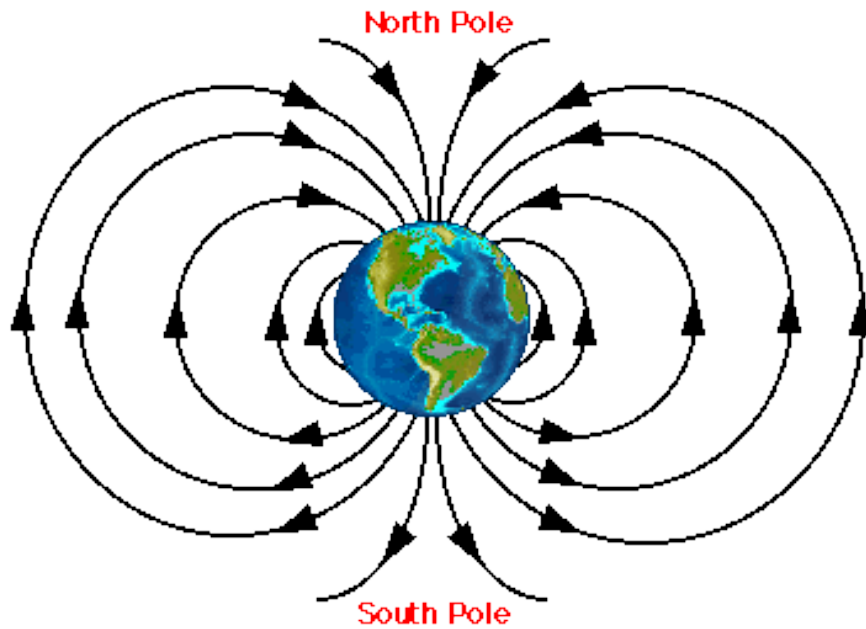
www.miniphysics.com

$$\nabla \times \underline{H} = \underline{J} + \frac{\partial \underline{D}}{\partial t}$$



**Magnetic field can be created
by current or changing electric
field (displacement current)**

The Earth's Magnetic Field

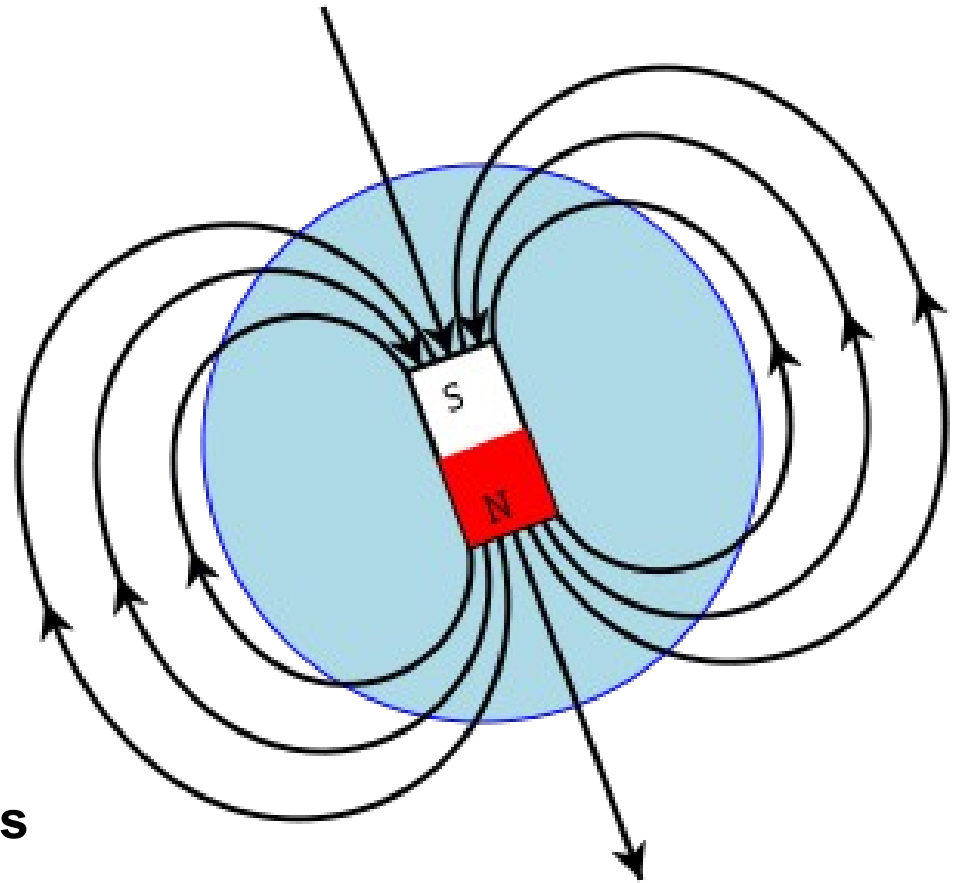


all-geo.org

Magnetic Gauss Law

en.wikipedia.org/wiki/Magnetic_field

$$\nabla \cdot \underline{B} = 0$$



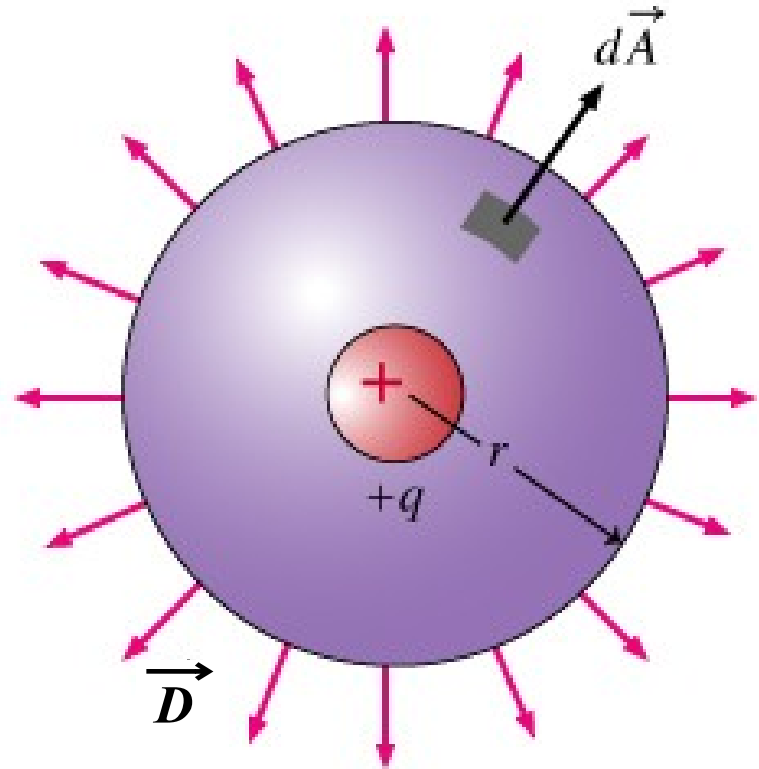
**Magnetic flux density is always
a divergence free vector field**

Electric Gauss Law

notendur.hi.is

The divergence of electric flux density is due to electric charge

$$\nabla \cdot \underline{D} = \rho_v$$



Law of Conservation of Electric Charge (Continuity Equation)

$$\nabla \times \underline{H} = \underline{J} + \frac{\partial \underline{D}}{\partial t}$$

$$\cancel{\nabla \cdot (\nabla \times \underline{H})} = \nabla \cdot \underline{J} + \nabla \cdot \left(\frac{\partial \underline{D}}{\partial t} \right)$$

$$0 = \nabla \cdot \underline{J} + \frac{\partial}{\partial t} (\nabla \cdot \underline{D})$$

Flow of electric
current out of volume
(per unit volume)

$$\nabla \cdot \underline{J} = - \frac{\partial \rho_v}{\partial t}$$

Rate of decrease of electric
charge (per unit volume)

Maxwell's Equations (cont.)

Time-Dependent

$$\nabla \times \underline{E} = - \frac{\partial \underline{B}}{\partial t} \quad \nabla \times \underline{H} = \underline{J} + \frac{\partial \underline{D}}{\partial t} \quad \nabla \cdot \underline{B} = 0 \quad \nabla \cdot \underline{D} = \rho_v$$

Time-Independent (Statics)

$$\nabla \times \underline{E} = 0 \quad \nabla \cdot \underline{D} = \rho_v$$

$$\nabla \times \underline{H} = \underline{J} \quad \nabla \cdot \underline{B} = 0$$

Decouples $\underline{E}$ and $\underline{H} \Rightarrow \underline{E}$ is a function of ρ_v and $\underline{H}$ is a function of $\underline{J}$

Maxwell's Equations (cont.)

Time-harmonic (phasor) domain

$$\frac{\partial}{\partial t} \rightarrow j\omega$$

$$\nabla \times \underline{\underline{E}} = -j\omega \underline{\underline{B}}$$

$$\nabla \times \underline{\underline{H}} = \underline{\underline{J}} + j\omega \underline{\underline{D}}$$

$$\nabla \cdot \underline{\underline{B}} = 0$$

$$\nabla \cdot \underline{\underline{D}} = \rho_v$$

Constitutive Relations

The characteristics of the media relate $\underline{D}$ to $\underline{E}$ and $\underline{H}$ to $\underline{B}$

Free Space

$$\underline{D} = \epsilon_0 \underline{E} \quad (\epsilon_0 = \text{permittivity})$$

$$\underline{B} = \mu_0 \underline{H} \quad (\mu_0 = \text{permeability})$$

$$\epsilon_0 \doteq 8.8541878 \times 10^{-12} \text{ [F/m]}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ [H/m] (exact)}$$

Constitutive Relations (cont.)

In a material medium:

$$\underline{\underline{D}} = \epsilon \underline{\underline{E}} \quad (\epsilon = \text{permittivity})$$

$$\underline{\underline{B}} = \mu \underline{\underline{H}} \quad (\mu = \text{permeability})$$

$$\epsilon = \epsilon_0 \epsilon_r \quad \epsilon_r = \text{relative permittivity}$$

$$\mu = \mu_0 \mu_r \quad \mu_r = \text{relative permeability}$$

Terminology

Properties of ϵ or μ

Variation	Independent of	Dependent on____
space	homogenous	inhomogeneous $\epsilon(x, y, z)$
frequency	non-dispersive	dispersive $\epsilon(\omega)$
time	stationary	non-stationary $\epsilon(t)$
field strength	linear	non-linear $\epsilon(E)$
direction of <u>E</u> or <u>H</u>	isotropic	anisotropic $\epsilon = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{bmatrix}$

Homogeneous and Inhomogeneous Media

homogeneous medium

ϵ & μ do n't change with space

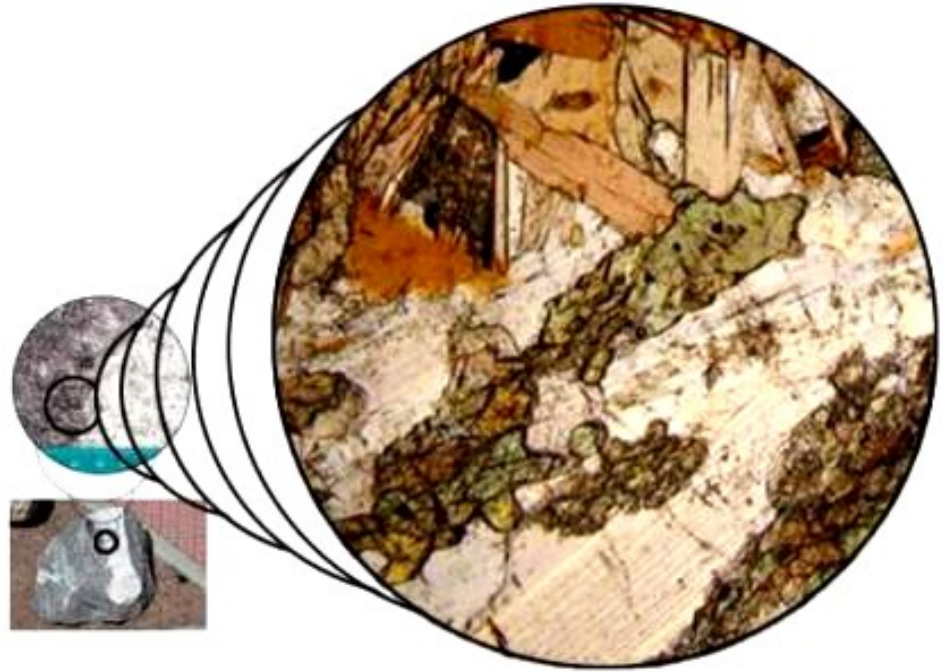


hamiltoncityschools.com

a bottle of pure water

inhomogeneous medium

$\epsilon(x, y, z)$ $\mu(x, y, z)$



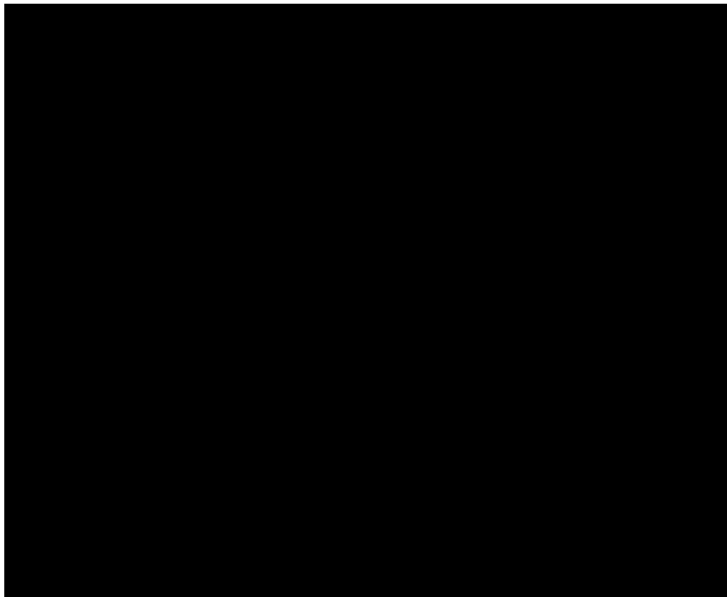
mckinneymuseum.appstate.edu

rock under microscope

Dispersive and Nondispersive Media

nondispersive medium

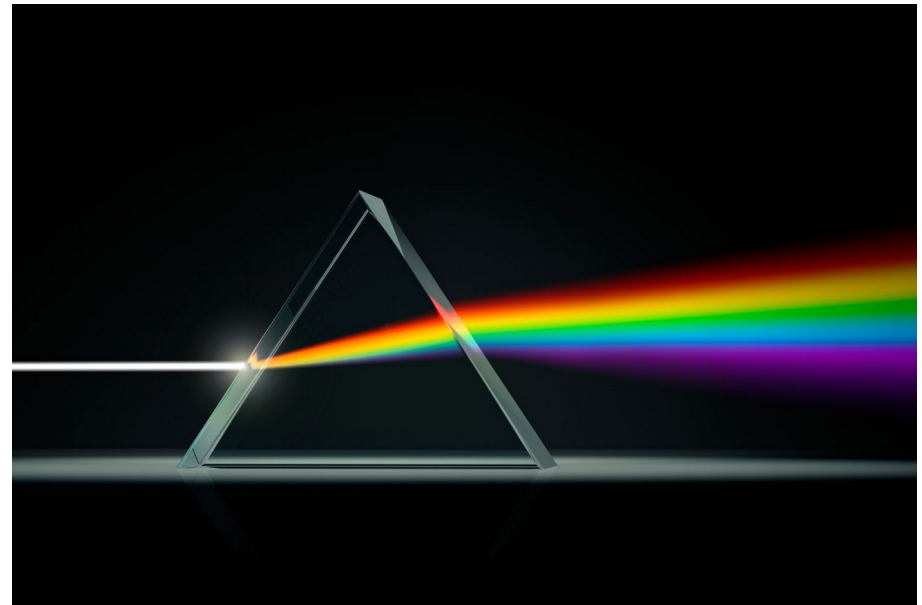
ϵ & μ do n't change with frequency



free space (vacuum)

dispersive medium

$\epsilon(\omega)$ $\mu(\omega)$



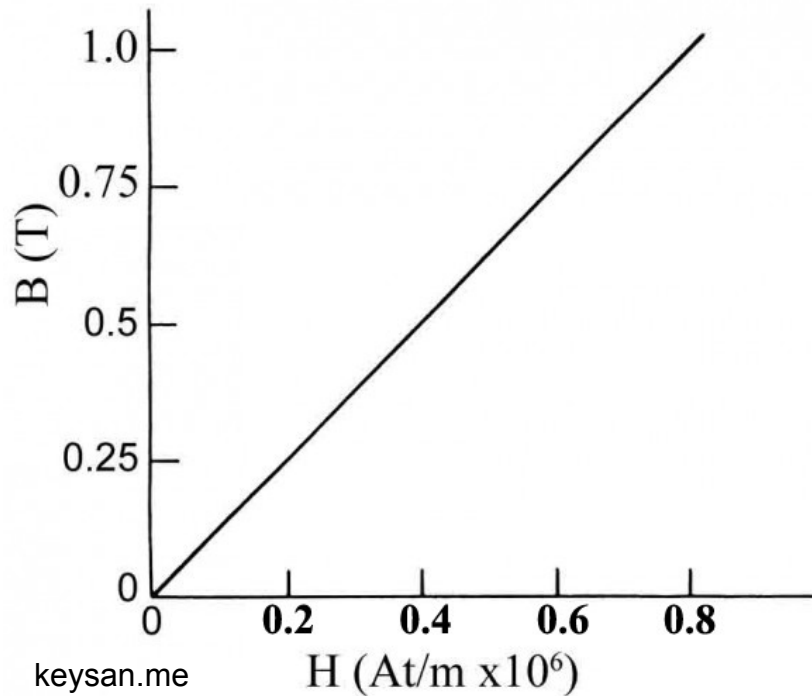
dkfindout.com

glass

Linear and Nonlinear Media

linear medium

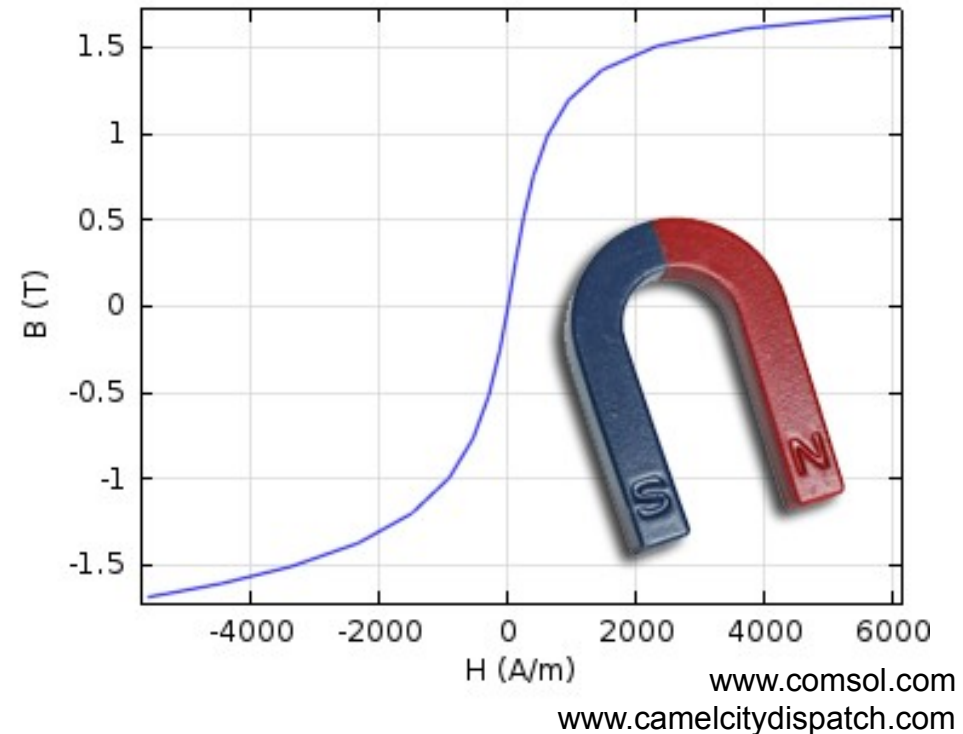
ϵ & μ do n't change with field strength



free space (vacuum)

nonlinear medium

$\epsilon(E)$ $\mu(H)$

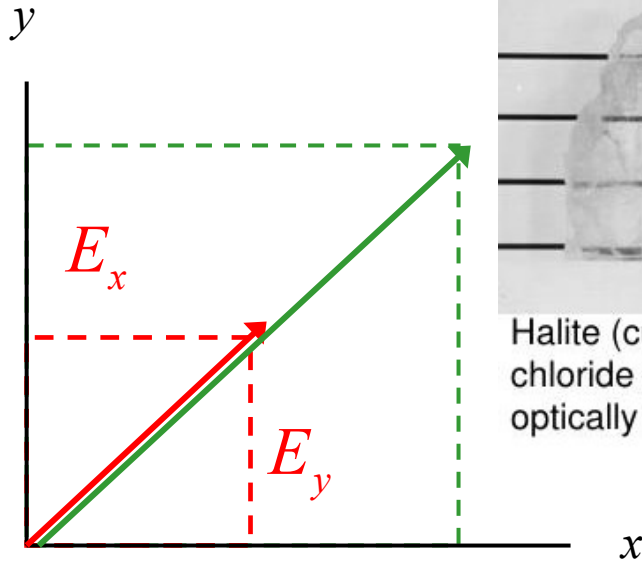


magnet / ferromagnetic material

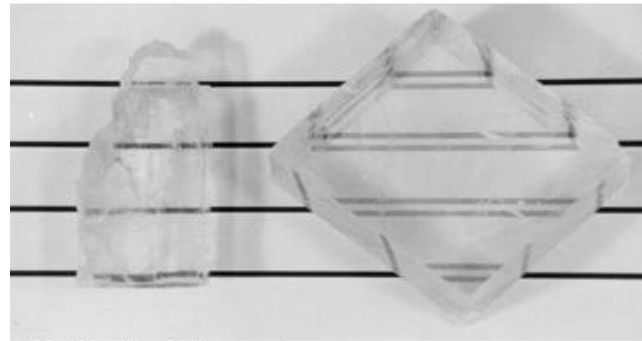
Isotropic and Anisotropic Media

isotropic medium

$\epsilon \wedge \mu$ are scalar numbers



free space (vacuum)



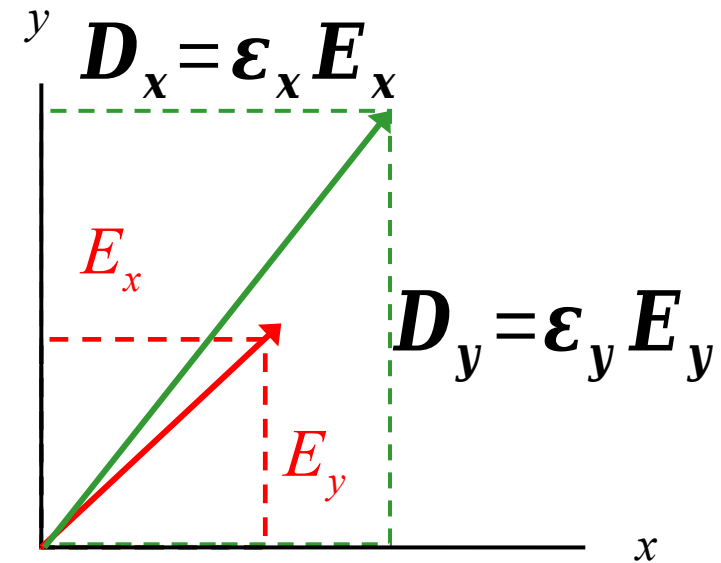
Halite (cubic sodium chloride crystal, optically isotropic)

Calcite (optically anisotropic)

www.slideserve.com

anisotropic medium

$$\epsilon = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{bmatrix} \quad \mu = \begin{bmatrix} \mu_{xx} & \mu_{xy} & \mu_{xz} \\ \mu_{yx} & \mu_{yy} & \mu_{yz} \\ \mu_{zx} & \mu_{zy} & \mu_{zz} \end{bmatrix}$$

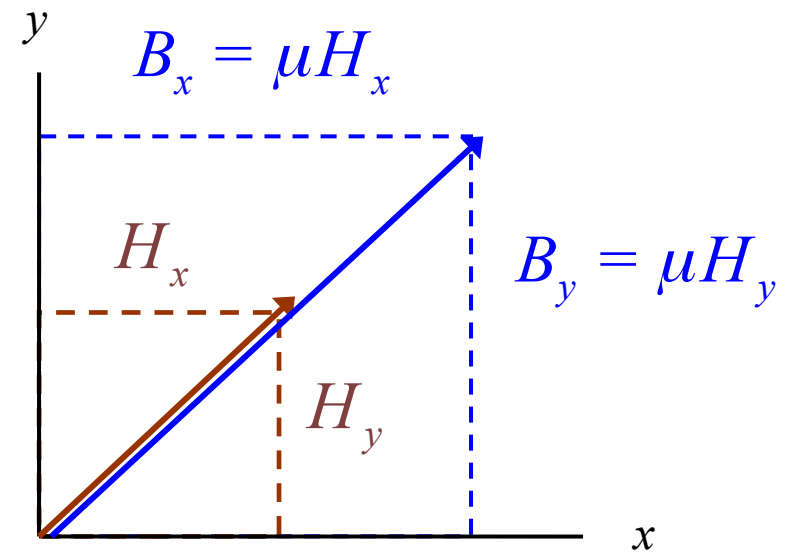
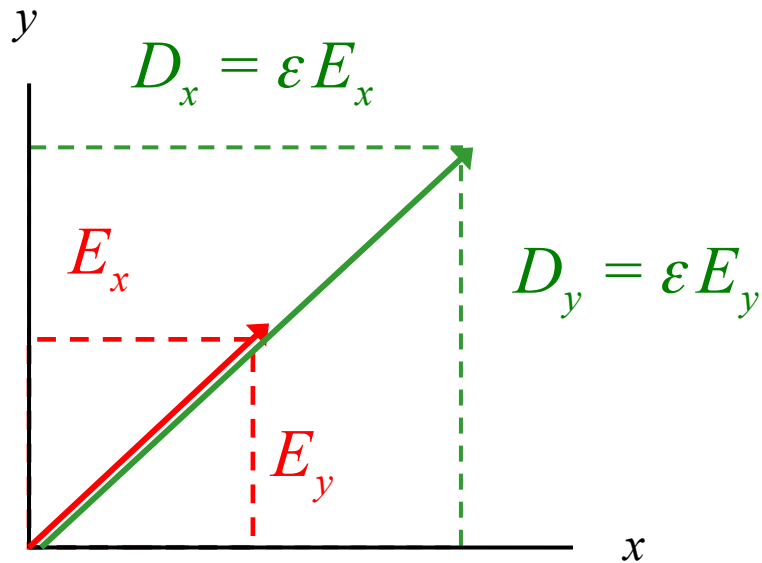


some type of orthotropic medium

Isotropic Materials

ε (and μ) are **scalar** quantities,
which means that $\underline{E} \parallel \underline{D}$ (and $\underline{H} \parallel \underline{B}$)

$$\underline{D} = \varepsilon \underline{E}$$
$$\underline{B} = \mu \underline{H}$$



Anisotropic Materials

ε (or μ) is a tensor (can be written as a matrix)

Example:

$$\left. \begin{aligned} D_x &= \varepsilon_x E_x \\ D_y &= \varepsilon_y E_y \\ D_z &= \varepsilon_z E_z \end{aligned} \right\} \quad \begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \begin{bmatrix} \varepsilon_x & 0 & 0 \\ 0 & \varepsilon_y & 0 \\ 0 & 0 & \varepsilon_z \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}$$

or $\underline{\underline{D}} = \underline{\underline{\varepsilon}} \cdot \underline{E}$

This results in $\underline{E}$ and $\underline{D}$ being **NOT** proportional to each other.

Anisotropic Materials (cont.)

Practical example: transversely isotropic medium

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \begin{bmatrix} \epsilon_h & 0 & 0 \\ 0 & \epsilon_h & 0 \\ 0 & 0 & \epsilon_v \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}$$

www.nap.edu



sedimentary rock

www.slb.com

